

# Maximum likelihood estimation

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# Logistics and reminders

- + HW 1 due tonight
- + Reminder: bank of 5 extension days -- can use 1 or 2 days on a homework assignment or project piece
- + HW 2 released, due next Friday
- + Project part 1 released (choosing a dataset + EDA)
  - + Project will be released in several pieces
  - + Part 1 due in 2 weeks
- + Coming up:
  - + Logistic regression assumptions and diagnostics
  - + Inference (hypothesis tests, confidence intervals)

## Warmup: maximum likelihood estimation

Work on the activity (handout), then we will discuss as a course

$$a^{b_1} a^{b_2} a^{b_3} = a^{b_1+b_2+b_3} = a^{\sum b_i}$$

## Warmup

Independent observations  $Y_1, \dots, Y_n$  with  
 $P(Y_i|\pi) = (1 - \pi)^{Y_i-1} \pi$ .

Likelihood function:

$$L(\pi) = P(Y_1|\pi)P(Y_2|\pi) \cdots P(Y_n|\pi) = \prod_{i=1}^n P(Y_i|\pi)$$

$$= \prod_{i=1}^n (1-\pi)^{Y_i-1} \pi$$

$$= \left[ \prod_{i=1}^n (1-\pi)^{Y_i-1} \right] \left[ \prod_{i=1}^n \pi \right]$$

$$= (1-\pi)^{Y_1-1 + Y_2-1 + Y_3-1 \cdots + Y_n-1} \pi^n$$

$$= (1-\pi)^{\sum_{i=1}^n (Y_i-1)} \pi^n = (1-\pi)^{\sum_{i=1}^n Y_i - n} \pi^n$$

# Warmup

$$L(\pi) = (1 - \pi)^{\sum_{i=1}^n Y_i - n} \pi^n$$

$$\log L(\pi) = \log \left( (1 - \pi)^{\sum_{i=1}^n Y_i - n} \right) + \log(\pi^n)$$

$$= (\sum_{i=1}^n Y_i - n) \log(1 - \pi) + n \log(\pi)$$

$$\frac{d}{d\pi} \log L(\pi) = \frac{(\sum_{i=1}^n Y_i - n)}{1 - \pi} (-1) + \frac{n}{\pi} \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \frac{n}{\pi} = \frac{(\sum_{i=1}^n Y_i - n)}{1 - \pi}$$

$$\Rightarrow \frac{1 - \pi}{\pi} = \frac{(\sum_{i=1}^n Y_i - n)}{n}$$

$$\Rightarrow \frac{1}{\pi} - 1 = \frac{\sum_{i=1}^n Y_i}{n} \quad \neq$$

$$\Rightarrow \pi = \frac{n}{\sum_{i=1}^n Y_i}$$

So  $\hat{\pi} = \frac{n}{\sum_{i=1}^n Y_i}$

# MLE for logistic regression

Suppose we observe independent samples  $(X_1, Y_1), \dots, (X_n, Y_n)$  with

$$\pi_i = \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}}$$

$$Y_i \sim \text{Bernoulli}(\pi_i) \quad \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 X_i$$

**Likelihood:**

$$\begin{aligned} Y_i \sim \text{Bernoulli}(\pi_i) &\Rightarrow P(Y_i | \pi_i) = \pi_i^{Y_i} (1 - \pi_i)^{1 - Y_i} \\ L(\beta_0, \beta_1) &= \prod_{i=1}^n \pi_i^{Y_i} (1 - \pi_i)^{1 - Y_i} \\ &= \prod_{i=1}^n \left( \frac{e^{\beta_0 + \beta_1 X_i}}{1 + e^{\beta_0 + \beta_1 X_i}} \right)^{Y_i} \left( \frac{1}{1 + e^{\beta_0 + \beta_1 X_i}} \right)^{1 - Y_i} \end{aligned}$$

calculus is hard, no closed form solution...

## Class activity, Part 2

Maximizing likelihood for a logistic regression model in R (linked on class website):

[https://sta214-f26.github.io/class\\_activities/ca\\_06\\_2.html](https://sta214-f26.github.io/class_activities/ca_06_2.html)

(You do not need to submit this part of the activity)